

Graduation Project Proposal

Project Title

SmartPick – An AI-Powered Web & Mobile Platform for Helping Egyptian Customers Choose the Right Smartphone

Problem Statement

In the Egyptian market, choosing a suitable smartphone has become increasingly difficult. There are dozens of brands, frequent model releases, confusing specifications, and large price differences. Most customers:

- Have a **limited budget**
- Care about **real usage** (gaming, social media, photography, work, battery life)
- Spend **hours watching YouTube reviews** and still end up confused
- Get stuck choosing between **2–3 similar phones** with no clear decision

Existing solutions either:

- Show raw specifications without guidance
- Promote sponsored content
- Do not consider **local prices, availability, or user priorities**

There is a clear need for an **intelligent, unbiased, usercentric system** that translates user needs into a clear recommendation.

Proposed Solution

SmartPick is an **AI-powered web and mobile platform** that helps Egyptian customers choose the best smartphone based on:

- Budget range
- Usage patterns
- Feature priorities
- Preferred brands
- Local market prices

Instead of forcing users to manually search, SmartPick **asks smart questions**, analyzes preferences, and provides:

-
- Ranked recommendations
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- Clear reasoning behind each recommendation
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- Dynamic comparison between any phones (without predefining them)

Core Features

1. Smart Recommendation Engine (AIDriven)

Users answer a short, guided questionnaire such as:

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- Budget (min / max)
-
- Main usage (gaming, camera, daily use, business, content creation)
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- Priority weights (battery, camera, performance, display, durability)
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- Preferred companies (Samsung, Xiaomi, Apple, Realme, etc.)
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- Dealbreakers (no AMOLED, no Snapdragon, no large phones, etc.)

The AI engine then:

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- Scores phones dynamically
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- Filters irrelevant options
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- Produces **ranked recommendations with explanations**

Example:

"This phone is recommended because it offers the best camera quality within your budget while maintaining strong battery life."

2. Dynamic Phone Comparison (No Predefined Limits)

Users can:

-
- Compare **any number of phones** dynamically
-
- Add or remove phones freely
-
- Compare based on:
 -
 - Realworld performance
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 - Camera quality (usecase based, not megapixels only)

- Battery endurance
- Value for money

The comparison is generated on the fly instead of static tables.

3. Egyptian Market Awareness

The system focuses on the **local Egyptian market** by:

- Using EGP prices
- Highlighting availability
- Accounting for local variants (RAM/storage differences)
- Detecting overpriced phones vs better alternatives

4. AI Explanation Layer

Instead of just saying "*Phone A is better than Phone B*", the platform explains:

- Why one phone fits the user better
- Tradeoffs clearly (e.g., better camera vs weaker battery)
- Which phone gives the **best value for the user, not on paper**

This builds trust and educational value.

5. Web & Mobile Experience

- Responsive Web App
- Mobile App (Android / iOS)
- Clean, Arabic-friendly UI
- Beginner-friendly language

Enhanced & Innovative Ideas (Added Value)

1. Confusion Resolver Mode

When users are stuck between 2–3 phones:

- The system asks **one or two decisive questions**

- Instantly gives a final recommendation

Example:

"If camera at night is more important than charging speed, choose Phone A. Otherwise, Phone B."

2. Usage Simulation Score

Instead of specs only, the system simulates usage:

- "Gaming Score"
- "Camera Lover Score"
- "Battery Endurance Score"

These scores are tailored to the user's usage profile.

3. Budget Stretch Advisor

If the user increases the budget slightly, the system shows:

- What extra value they gain
- Whether the upgrade is worth it or not

Example:

"Adding 1,500 EGP gives you 25% better camera quality and faster charging."

4. AI Chat Assistant

An integrated assistant that allows users to ask questions like:

- "Is this phone good for university and light gaming?"
- "What will I lose if I choose this cheaper phone?"
- "Which phone ages better over 3 years?"

5. Learning & Trust Layer

- Explains technical terms in simple Arabic
- Educates users gradually
- Helps users become smarter buyers

System Architecture (High Level)

- • **Frontend:** Web & Mobile UI
- • **Backend:** API for data, recommendations, and comparisons
- • **AI Layer:**
 - • Rulebased scoring (initial phase)
 - • Machine learning model (future enhancement)
- • **Database:**
 - • Phone specifications
 - • Market prices
 - • User preference profiles

Technologies (Suggested)

- • Frontend: React / React Native / Flutter
- • Backend: Node.js / Express or Django
- • Database: MongoDB / PostgreSQL
- • AI: Python (ML models), rulebased decision system
- • Hosting: Cloud deployment

Target Users

- • Egyptian students
- • Budgetconscious buyers
- • Nontechnical users
- • Anyone confused by smartphone choices

Project Impact

- • Saves users **time, money, and confusion**
- • Promotes **smart spending instead of brand hype**

- • Bridges the gap between technical specs and real needs
- • Has strong potential as a **commercial and scalable product**

Future Enhancements

- • User reviews and feedback loop
- • Price tracking & alerts
- • Usedphone recommendations
- • Accessories suggestions (power banks, cases)
- • Expansion to other electronics (laptops, tablets)

Conclusion

SmartPick aims to transform how Egyptian customers choose smartphones by shifting from **specdriven confusion** to **AIguided clarity**. The project combines AI, web/mobile development, UX design, and realworld problem solving, making it a strong and practical graduation project.